

Private Pilot (ASEL) Oral Exam Question Bank

Organized by ACS Area of Operation and Task, based on FAA-S-ACS-6C (effective May 31, 2024), and tailored to N8821U (Cessna 172F) at Brazos Valley Flight Services.

How to use this guide

This guide mirrors the structure a Designated Pilot Examiner (DPE) actually tests from: the Private Pilot – Airplane Airman Certification Standards (ACS). Every Area of Operation and Task is labeled with its official ACS numbering (e.g., Area I, Task A) so you can cross-reference this guide directly against the ACS itself. Questions are drawn from the ACS Knowledge and Risk Management elements — the parts the oral exam actually probes — supplemented with questions DPEs and CFIs most frequently report asking on real checkrides. Every answer carries at least one source reference so you can go verify it yourself rather than trust rote memorization — applicants who can locate an answer in the regulations or handbooks pass more easily than those who only memorized a line.

This guide is written for N8821U specifically (a 1965 Cessna 172F, Continental O-300-C, fixed-pitch prop, generator-to-alternator conversion). Confirm every aircraft-specific answer against N8821U's actual POH/AFM and checklist before your checkride — an examiner will fail you for reciting the wrong airplane's numbers, and this guide can't see modifications made after it was written.

Two tips before you start studying: - **Answer the question asked, then stop.** Volunteering extra information beyond what was asked often invites a harder follow-up question that wasn't coming. If the examiner wants more detail, they'll ask for it. - **Speak in your own words, not acronyms.** An examiner can tell when an answer is recited from a mnemonic rather than understood — be ready to explain the substance behind any acronym you use.

Reference key

Abbreviation	Source
FAR	Title 14 Code of Federal Regulations (14 CFR) — cited by part.section
AIM	Aeronautical Information Manual
PHAK	Pilot's Handbook of Aeronautical Knowledge, FAA-H-8083-25
AFH	Airplane Flying Handbook, FAA-H-8083-3
AWH	Aviation Weather Handbook, FAA-H-8083-28
RMH	Risk Management Handbook, FAA-H-8083-2
IFH	Instrument Flying Handbook, FAA-H-8083-15
ACS	Private Pilot – Airplane ACS, FAA-S-ACS-6C

Aircraft-specific reference — N8821U (Cessna 172F)

Airframe/engine

Item	N8821U
Engine	Continental O-300-C — 6-cylinder, horizontally opposed, air-cooled, direct-drive, normally aspirated, carbureted
Rated power	145 BHP at 2,700 RPM
Propeller	Fixed-pitch, McCauley 1C172/EM7653, 76 in. diameter
Magnetos	Dual; check at 1,600 RPM, max 75 RPM differential/drop per side
Landing gear	Fixed tricycle; oleo-pneumatic nose strut with torque links
Electrical	14-volt DC, engine-driven alternator (converted from the original generator), low-voltage/alternator warning light; 12-volt battery on the left forward firewall
Hydraulic fluid	MIL-H-5606 (brakes/nose strut servicing)
Anti-icing/de-icing	None noted in this POH — no pitot heat listed; not approved for flight into known icing

Fuel system: Gravity-fed (no engine-driven or electric fuel pump) from two wing tanks through a selector valve (OFF/LEFT/RIGHT/BOTH) and strainer to the carburetor. Minimum fuel grade per this 1965 POH is 80/87 — not produced commercially anymore; 100LL is the FAA/Continental-approved substitute (a higher-octane fuel is always acceptable for a lower-octane minimum). Capacity: 19.5 gal/tank total (39 gal total); usable fuel 18.0 gal/tank in all flight conditions, up to 19.0 gal/tank in level flight only (36–38 gal usable total); 1.5 gal/tank unusable under maneuvering conditions. Oil capacity: 8 qt (do not operate below 6 qt).

V-speeds (N8821U checklist, MPH — this airframe is placarded/operated in MPH, not knots)

Speed	Value	Meaning
V _{so}	49 MPH	Stall speed, landing configuration (full flaps)
V _{s1}	57 MPH	Stall speed, flaps up
V _r	60 MPH	Rotation speed
V _x (50 ft obstacle)	65 MPH	Best angle of climb, obstacle clearance
V _x	70 MPH	Best angle of climb, general
V _g	80 MPH	Best glide
V _y	80 MPH	Best rate of climb (sea level)
V _{fe}	100 MPH	Maximum flap extended speed
V _a	122 MPH	Maneuvering speed (max gross weight, 2,300 lb)
V _{no}	140 MPH	Maximum structural cruising speed (top of green arc)
V _{ne}	174 MPH	Never-exceed speed (red line)

Weight, limits, and performance: Gross weight 2,300 lb (Utility category limited to 2,000 lb). Empty weight (typical) ≈1,260 lb. Baggage limit 120 lb. Load factors: +3.8/−1.52 G flaps up, +3.5 G flaps down (Normal); +4.4/−1.76 G flaps up (Utility, 2,000 lb max, no rear-seat/baggage occupant). Service ceiling 13,100 ft. Flap operating range (white arc) 52–100 MPH; normal operating range (green arc) 59–140 MPH; caution range (yellow arc) 140–174 MPH. Full-flap slips are prohibited (POH warning — pitch-down tendency at certain airspeed/sideslip combinations with 40° flaps).

References: Cessna 172/Skyhawk Owner's Manual (1965), Sections I–V; N8821U checklist, BVFS

I. Preflight Preparation

A. Pilot Qualifications

1. What are the requirements to obtain a private pilot certificate? **A:** Be at least 17 years old; be able to read, speak, write, and understand English (or receive a limiting endorsement); hold at least a student pilot certificate; pass the private pilot aeronautical knowledge (written) test; receive the required flight training and logbook endorsements from an authorized instructor; meet the aeronautical experience requirements of 61.109; pass the practical test (checkride) on the areas of operation. *Ref: 14 CFR 61.103, 61.105, 61.107, 61.109; ACS Area I.A*
2. What aeronautical experience is required under Part 61? **A:** At least 40 hours of total flight time, including 20 hours of flight training from an authorized instructor (3 hours cross-country, 3 hours night — including one 100+ NM cross-country and 10 night full-stop takeoffs/landings, 3 hours instrument, 3 hours checkride prep within the preceding 2 calendar months) and 10 hours of solo flight time (5 hours solo cross-country, including one 150 NM solo cross-country with full-stop landings at three points and one segment over 50 NM, plus 3 takeoffs/landings to a full stop at a towered airport). *Ref: 14 CFR 61.109*
3. At what age can a student pilot solo, and at what age can you take the checkride? **A:** Solo: 16 years old (14 for gliders/balloons). Practical test/certificate issuance: 17 years old (16 for gliders/balloons). *Ref: 14 CFR 61.83, 61.87, 61.103*
4. What makes a pilot current to carry passengers? **A:** Within the preceding 90 days, 3 takeoffs and 3 landings in an aircraft of the same category, class, and type (if required). At night, those landings must be to a full stop (touch-and-goes don't count) between one hour after sunset and one hour before sunrise — meeting the night requirement also satisfies day currency. *Ref: 14 CFR 61.57(a); AIM 8-1-1*
5. What are the different definitions of "night" used in aviation, and why does it matter? **A:** Sunset to sunrise — the period position/anti-collision lights are required (91.209). End of evening civil twilight to beginning of morning civil twilight — the period logged as night flight time (61.51(b)(3)(i)). One hour after sunset to one hour before sunrise — the window for passenger-carrying takeoff/landing currency (61.57(b)). These three windows are nested and don't line up exactly — a favorite oral "gotcha," so know which definition governs which requirement. *Ref: 14 CFR 91.209, 61.51(b)(3)(i), 61.57(b)*
6. What is the difference between being current and being proficient? **A:** Currency means you've met the legal minimums to exercise a privilege (flight review, passenger-carrying currency, valid medical). Proficiency means you actually possess the skill and recent practice to perform the flight safely — a pilot

can be legally current and still not safe for a given mission (e.g., first actual IMC approach in 8 months). Personal minimums exist to close this gap. *Ref: PHAK Ch. 2; AC 61-98*

7. How do you keep your pilot certificate legally current to act as PIC? **A:** Complete a flight review (1 hour ground, 1 hour flight training with an authorized instructor) every 24 calendar months, with a logbook endorsement. A new certificate/rating, a proficiency check, or completing specific FAA WINGS program activity can substitute. *Ref: 14 CFR 61.56*

8. What are the privileges and limitations of a private pilot certificate? **A:** Privileges: act as PIC for any purpose not for compensation or hire; carry passengers and property; share operating expenses pro rata with passengers; act as PIC of a charitable/nonprofit/community-event flight under specific conditions; act as PIC for nonprofit search-and-rescue; act as an aircraft salesperson with at least 200 hours logged. Limitations: may not be paid for acting as PIC, or carry passengers/property for compensation or hire, outside those narrow exceptions. *Ref: 14 CFR 61.113*

9. What does "pro rata" mean in the context of shared flight expenses? **A:** An equal share among all occupants who could act as PIC and passengers, of the operating expenses of a flight — fuel, oil, aircraft rental, landing/tie-down fees. Splitting costs with one passenger is half each; the pilot can never be reimbursed for more than an equal share. *Ref: 14 CFR 61.113(c)*

10. You own a business — can you fly your own plane on the clock to go look at something for it, such as scouting a vehicle in another state? **A:** Yes, as long as you aren't being paid specifically for the fact that you're flying — the flight has to be "incidental to the business." You could have driven or taken a commercial flight instead; the business purpose is what's being compensated, not the piloting. *Ref: 14 CFR 61.113*

11. A friend offers to pay you to fly him somewhere to pick up a car he bought. Can you accept payment for that flight? **A:** No — that's compensation for the flight itself, which a private pilot can't accept, unless you had your own independent reason ("common purpose") to make that same trip anyway, in which case you can share pro-rata operating expenses (never more than an equal share) with him. *Ref: 14 CFR 61.113(c)*

12. Can you post on social media asking if anyone wants to fly with you somewhere and split gas? **A:** No — that's "holding out," publicly offering your flight to whoever wants it, which starts to look like common carriage. Sharing pro-rata expenses is only legal with someone you already had a reason to invite, not an open solicitation. *Ref: 14 CFR 61.113(c); AC 61-142*

13. What are the classes of medical certificates, and what are they generally used for? **A:** 1st class — required for ATP privileges. 2nd class — required for commercial pilot privileges (compensated flying). 3rd class — sufficient for private, recreational, and student pilot privileges. A higher class also satisfies any lower class while its own privileges remain valid. *Ref: 14 CFR 61.23*

14. How long is your medical certificate valid, and how do the classes downgrade? **A:** Under age 40 at exam: a 3rd-class medical is valid 60 calendar months; a 1st- or 2nd-class medical carries its higher-class privileges for 12 months, then downgrades to 3rd-class privileges through the end of the 60th month. Age 40 or older: a 3rd-class medical is valid 24 months; a 1st-class medical gives 1st-class privileges for 6 months, then 2nd-class for the next 6, then 3rd-class through month 24; a 2nd-class medical gives 2nd-class privileges for 12 months, then 3rd-class through month 24. A medical always expires on the last day of its specified expiration month. *Ref: 14 CFR 61.23(d)*

15. What is BasicMed, and what are its privileges and limitations? **A:** An alternative to holding an FAA medical certificate. Requirements: held a valid FAA medical certificate at some point after July 14, 2006; most recent medical application wasn't denied/revoked/suspended; a comprehensive physical exam with a state-licensed physician every 48 months using the FAA's checklist; a free FAA-approved online medical education course every 24 months, with proof of completion carried. Limitations: aircraft authorized for takeoff with 6 or fewer occupants and max certificated takeoff weight of 12,500 lbs or less; VFR or IFR within the US at or below 18,000 ft MSL, no faster than 250 KIAS; no more than 6 passengers; not for compensation or hire. *Ref: 14 CFR Part 68; AC 68-1A*
16. What documents must a pilot have in their possession when acting as PIC? **A:** Government-issued photo ID, pilot certificate, and (unless operating under BasicMed or as a glider/balloon pilot exempt from the requirement) a current medical certificate. *Ref: 14 CFR 61.3*
17. What are the alcohol and drug rules for pilots? **A:** May not act as a crewmember within 8 hours of consuming alcohol ("8 hours bottle to throttle"), while under the influence, with a blood alcohol concentration of 0.04 or greater, or while using any drug that affects your faculties in a way contrary to safety. May not allow anyone who appears intoxicated or impaired aboard, except a medical patient under proper care. *Ref: 14 CFR 91.17*
18. What is IMSAFE, and when do you use it? **A:** A personal self-assessment checklist run before every flight: Illness, Medication, Stress, Alcohol, Fatigue, Eating/hydration. If any item is unsatisfactory, don't fly. *Ref: PHAK Ch. 2; RMH Ch. 4*
19. What must you do if you develop a medical condition that no longer meets your medical certificate's standards? **A:** You may not act as PIC (or any required crewmember) while you know, or have reason to know, of a medical deficiency, or while taking medication that makes you unable to meet the certificate's standards. Self-grounding is the pilot's own responsibility, medical certificate in hand or not. *Ref: 14 CFR 61.53*
20. What happens if you fail a portion of the practical test? **A:** The DPE issues a Notice of Disapproval listing the failed Areas of Operation. You may request a partial retest of just those items, typically within 60 days, once retrained and endorsed by an instructor. If more than 60 days pass, a full retest is required. *Ref: 14 CFR 61.49; FAA Order 8900.2*
21. What are the three possible outcomes of a checkride, and what specifically distinguishes "incomplete" from "unsatisfactory"? **A:** Satisfactory (consistent performance within ACS standards — not flawless, but correcting deviations appropriately), Unsatisfactory (consistently outside standards, or an action that jeopardizes safety), and Incomplete (the test can't be finished for a reason outside the applicant's knowledge/skill — weather, an aircraft airworthiness issue, sudden illness). Being unprepared, or simply not knowing an answer, is never grounds for "incomplete" — that's the applicant's own responsibility. *Ref: 14 CFR 61.43(b)*

B. Airworthiness Requirements

22. What documents are required to be in the aircraft? **A:** Mnemonic A.R.R.O.W. — Airworthiness certificate (never expires on its own, as long as the aircraft meets its type design and is properly

maintained), Registration certificate (valid 7 years from issuance, or until sold/revoked/destroyed; non-transferable), Radio station license (only required for international operations), Operating limitations (placards, markings, POH/AFM limitations section), Weight and balance data (current report and equipment list). *Ref: 14 CFR 91.203, 47.40, 91.9; AIM 4-1-1*

23. What does "airworthy" mean, and who determines it? **A:** The aircraft conforms to its type certificate (or approved alterations) and is in condition for safe operation. The pilot in command makes the final determination before every flight, based on required inspections, ADs, and a satisfactory preflight. *Ref: 14 CFR 91.7; PHAK Ch. 8*

24. Where must the airworthiness certificate be displayed? **A:** In a conspicuous place in the aircraft so it's legible to passengers and crew — typically the cabin or cockpit entrance. *Ref: 14 CFR 91.203(b)*

25. What inspections does the aircraft require? **A:** Mnemonic AV1ATES — Annual (every 12 calendar months, by an A&P with Inspection Authorization), VOR check (every 30 days, only if flown IFR under VOR navigation), 100-hour (every 100 hours, only if used for hire or flight instruction for hire), ADs (complied with as scheduled), Transponder (every 24 calendar months), ELT (every 12 calendar months, and after 1 cumulative hour of use or 50% battery life expired), Static system/altimeter/pitot (every 24 calendar months, required for IFR only). *Ref: 14 CFR 91.409, 91.411, 91.413, 91.207, 91.171*

26. Can an annual or 100-hour inspection be exceeded (flown over)? **A:** An overdue annual grounds the aircraft; it can only be flown with a special flight permit (ferry permit) from the FSDO, typically to reach a shop. A 100-hour inspection may be exceeded by up to 10 hours only while en route to the inspection, and that extra time counts against the next 100-hour interval. An annual satisfies the 100-hour requirement, but not vice versa. *Ref: 14 CFR 91.405, 91.409(a)(2)*

27. Who can perform preventive maintenance on your aircraft? **A:** A Part 61 pilot certificate holder (other than sport pilot, which has its own narrower list) may perform preventive maintenance on any aircraft they own or operate — not just aircraft they own outright — as long as it isn't operated under Part 121, 129, or 135; this applies to a student pilot certificate holder too. The roughly 30 items on Part 43 Appendix A(c) (oil/filter changes, servicing tires, replacing bulbs/fuses, spark plug cleaning, etc.) don't require any additional certificate, but the work must be logged. Exception: Appendix A(c)(30) requires a "certificate of competency," but only for primary-category aircraft — a narrow certification category distinct from Normal/Utility/Acrobat/Transport. N8821U is certificated Standard/Normal-Utility under Type Certificate 3A12, not primary category, so this exception doesn't apply to it. *Ref: 14 CFR 43.3(g), 43.7(f), 43.9; Part 43 Appendix A(c), A(c)(30); 14 CFR 21.24*

28. What equipment is required for VFR day flight? **A:** Mnemonic A TOMATO FLAMES — Airspeed indicator, Tachometer (per engine), Oil pressure gauge, Manifold pressure gauge (per altitude engine), Altimeter, Temperature gauge (per liquid-cooled engine), Oil temperature gauge (per air-cooled engine), Fuel gauge (per tank), Landing gear position indicator (retractable gear), Anti-collision lights, Magnetic compass, ELT, Seat belts (and shoulder harnesses for required crew). *Ref: 14 CFR 91.205(b)*

29. What additional equipment is required for VFR night flight? **A:** Mnemonic FLAPS, added to the day list — Fuses (a spare set, or three of each kind), Landing light (only if operated for hire), Anti-collision lights, Position (navigation) lights, Source of electrical power adequate for the equipment used. *Ref: 14 CFR 91.205(c)*

30. What would you do if you discovered inoperative equipment before flight? **A:** Determine whether the item is required by the type certificate/equipment list, an AD, 91.205, or the kind of operation being conducted. If not, it may be placarded "Inoperative" and deactivated under 91.213(d) (absent an approved MEL), provided a certificated mechanic determines it's safe to fly without it. If it is required, the aircraft is unairworthy until repaired, replaced, or an appropriate deferral applies. *Ref: 14 CFR 91.213*
31. What is an MEL, and what is a KOEL? **A:** An MEL (Minimum Equipment List) is an FAA-approved, aircraft-specific list of equipment that may be inoperative for dispatch — most single-engine trainers don't have one, so 91.213(d) governs instead. A KOEL (Kinds of Operation Equipment List), in the POH limitations section, lists equipment required for specific kinds of operations (day VFR, night VFR, IFR). *Ref: 14 CFR 91.213; AC 91-67*
32. When must a maintenance logbook entry be made? **A:** Immediately after any inspection, maintenance, preventive maintenance, alteration, or rebuild, and before the aircraft is approved for return to service — describing the work performed and signed by the person who performed it. *Ref: 14 CFR 43.9, 43.11*
33. What is a Special Flight Permit (ferry permit), and when would you need one? **A:** An FAA authorization to fly an aircraft that doesn't currently meet airworthiness requirements — e.g., overdue for annual, or needing a repair not available locally — to a place where the work can be done, or for delivery/export/relocation. Issued by the FSDO with specific conditions and a limited route/duration. *Ref: 14 CFR 21.197, 21.199*
34. What are the standard airworthiness certificate categories, and which does your aircraft hold? **A:** Standard categories: Normal, Utility, Acrobatic, Transport, Commuter. Special categories: Primary, Restricted, Limited, Light-Sport, Experimental. N8821U holds a Standard airworthiness certificate in the Normal/Utility category under Type Certificate 3A12. *Ref: 14 CFR 21.175, 21.184; Owner's Manual*

C. Weather Information

35. How can you obtain a weather briefing before flight? **A:** 1800wxbrief.com, the ForeFlight/Garmin Pilot app (Leidos Flight Service data), AviationWeather.gov, or calling Flight Service (1-800-WX-BRIEF). *Ref: AIM 7-1-2, 7-1-4*
36. What are the types of weather briefings, and when is each appropriate? **A:** Standard briefing — complete, used for the first briefing of the day or after 6+ hours since the last one. Abbreviated briefing — updates a previous one or covers specific information only. Outlook briefing — used when departure is 6+ hours away, for planning only; must be followed by a standard or abbreviated briefing closer to departure. *Ref: AIM 7-1-4*
37. What is the difference between a METAR and a TAF? **A:** A METAR is a routine, hourly (or special/SPECI when conditions change significantly) surface observation of actual current weather. A TAF is a forecast, issued roughly every 6 hours, valid 24–30 hours, describing expected conditions within about 5 statute miles of the airport. *Ref: AIM 7-1-27, 7-1-28; AWH Ch. 12–13*
38. Decode this METAR: KXXX 151753Z 18012G20KT 6SM -RA BKN025 OVC040 18/16 A2985 RMK A02 **A:** Station KXXX, observed the 15th at 1753Z, wind 180° at 12 knots gusting 20, visibility 6 statute

miles, light rain, broken clouds at 2,500 ft AGL, overcast at 4,000 ft AGL, temperature 18°C, dew point 16°C, altimeter 29.85 inHg, remark: automated station with precipitation discriminator. *Ref: AIM 7-1-27; AWH Ch. 12*

39. Decode this TAF excerpt: TAF KXXX 151720Z 1518/1624 20010KT P6SM FEW040 FM152200 22015G22KT 4SM -SHRA BKN020 **A:** Forecast for KXXX, issued the 15th at 1720Z, valid from the 15th at 1800Z to the 16th at 2400Z: initially wind 200° at 10 kt, visibility greater than 6 SM, few clouds at 4,000 ft; from 2200Z on the 15th, wind 220° at 15 kt gusting 22, visibility 4 SM, light rain showers, broken clouds at 2,000 ft. *Ref: AIM 7-1-28; AWH Ch. 13*

40. What is a PIREP, and why is it valuable? **A:** A pilot report of actual in-flight conditions (turbulence, icing, cloud tops/bases, visibility) relayed to ATC or Flight Service. It's the only "ground truth" source of current en route conditions — forecasts are inferential, a PIREP is an eyewitness account. *Ref: AIM 7-1-20; FAA AC 00-45*

41. What is a SIGMET, and how long is it valid? **A:** Significant Meteorological Information — a weather advisory affecting all aircraft, covering severe icing, severe/extreme turbulence, dust/sandstorms below 3 SM visibility, and volcanic ash. Valid up to 4 hours (6 for hurricanes). *Ref: AIM 7-1-6*

42. What is a Convective SIGMET? **A:** An advisory for convective weather hazardous to all aircraft: severe thunderstorm lines, embedded thunderstorms, thunderstorm areas affecting 40%+ of a region, or a single severe/embedded thunderstorm with tornadoes, hail $\geq 3/4$ ", or wind gusts ≥ 50 knots. Valid 2 hours. *Ref: AIM 7-1-7*

43. What are AIRMETs, and what are the three types? **A:** Weather advisories of concern particularly to light aircraft, less severe than SIGMET criteria, issued every 6 hours. Sierra — IFR conditions/extensive mountain obscuration. Tango — moderate turbulence, sustained winds ≥ 30 kt, or low-level wind shear. Zulu — moderate icing and freezing levels. *Ref: AIM 7-1-6*

44. What are the categories of turbulence, and how are they defined? **A:** Light, moderate, severe, and extreme, defined by the effect on the aircraft (momentary bumps vs. large abrupt changes vs. occupants forced against restraints vs. momentarily out of control), not by a specific wind value. PIREPs are the most reliable source, since forecasts only outline areas of probability. *Ref: AIM 7-1-24; AWH Ch. 11*

45. What are the types of structural icing, and how do they differ? **A:** Rime ice — rough, milky, opaque, forms quickly from small supercooled droplets, typical in stratiform clouds. Clear ice — smooth, hard, transparent, most dangerous/hardest to remove, from larger supercooled droplets that spread before freezing. Mixed ice — a rough combination of both. Structural icing always requires visible moisture and a temperature at or below freezing at the airframe surface. *Ref: AWH Ch. 14; PHAK Ch. 12*

46. What are the VFR weather minimums? **A:** Class B: 3 SM, clear of clouds. Class C/D/E below 10,000 ft MSL: 3 SM, 500 below/1,000 above/2,000 horizontal. Class E at/above 10,000 ft MSL: 5 SM, 1,000/1,000/1 SM. Class G $\leq 1,200$ ft AGL: 1 SM day (clear of clouds) / 3 SM night (500/1,000/2,000). Class G above 1,200 ft AGL, below 10,000 ft MSL: 1 SM day / 3 SM night, 500/1,000/2,000. Class G above 1,200 ft AGL, at/above 10,000 ft MSL: 5 SM, 1,000/1,000/1 SM. *Ref: 14 CFR 91.155*

47. What are the different types of fog, and how does each form? **A:** Radiation fog — forms over land on clear, calm nights as the surface radiates heat away, common in valleys, usually burns off after sunrise.

Advection fog — warm, moist air moves over a cooler surface (coasts), can persist until wind/air mass changes. Upslope fog — moist stable air mechanically lifted up rising terrain. Steam fog — cold, dry air moves over warm water. Precipitation-induced (frontal) fog — warm rain falls through cooler air, raising its dew point to saturation; associated with warm fronts and slow-moving cold fronts. *Ref: AWH Ch. 15*

48. How do high- and low-pressure systems behave, and why does it matter to a VFR pilot? **A:** Northern Hemisphere: high pressure circulates clockwise and outward (fair weather); low pressure circulates counterclockwise and inward (clouds/precipitation). "Flying from high to low, look out below" — flying toward lower pressure or warmer temperature without resetting the altimeter causes it to read higher than true altitude, meaning you're actually lower than indicated. *Ref: PHAK Ch. 12; AWH Ch. 5*

49. What is an isobar, and what does isobar spacing tell you? **A:** A line connecting points of equal atmospheric pressure. Closely spaced isobars indicate a steep pressure gradient and stronger winds; widely spaced isobars indicate light winds. *Ref: AWH Ch. 5*

50. Describe a warm front and a cold front. **A:** Warm front — warm air rides up over retreating cooler air on a shallow slope (~1:200), producing widespread stratiform clouds and steady precipitation well ahead of the surface front, moving slowly (~10–25 mph). Cold front — cold, dense air wedges under and lifts warm air abruptly on a steep slope, producing cumuliform clouds, showery/heavy precipitation, turbulence, and sometimes thunderstorms, moving fast (up to ~60 mph) with rapid clearing behind. *Ref: AWH Ch. 5; PHAK Ch. 12*

51. What is an occluded front, and what is a stationary front? **A:** Occluded — a faster cold front catches a slower warm front; expect warm-front weather followed by cold-front weather in sequence. Stationary — two air masses meet with little relative motion; associated weather can linger over the same area for days. *Ref: AWH Ch. 5*

52. What is a trough? **A:** An elongated area of relatively low atmospheric pressure, often associated with a wind shift and a line of cloudiness/precipitation, without necessarily having a fully developed frontal boundary. *Ref: AWH Ch. 5*

53. What three ingredients are required for a thunderstorm to form? **A:** Sufficient moisture, an unstable air mass (a lapse rate letting a rising parcel stay warmer than its surroundings), and a lifting force (frontal, orographic, or surface heating/convection). *Ref: AWH Ch. 13; PHAK Ch. 12*

54. What happens when temperature and dew point converge? **A:** The air approaches saturation, favoring fog, low clouds, and haze, and increasing the likelihood of IMC. A spread of 2–3°C or less that's narrowing is a classic warning sign on a METAR/TAF. *Ref: AWH Ch. 4; PHAK Ch. 12*

55. What are personal minimums, and how do you set them? **A:** Self-imposed limits stricter than legal/regulatory minimums, set in advance rather than in the moment, using a tool like PAVE (Pilot, Aircraft, enVironment, External pressures) to account for actual experience, recency, and the mission at hand. Example: "I will not depart VFR if the forecast ceiling is below 2,500 ft AGL or visibility below 5 SM." Knowing your minimums is just a number on paper — living by them under real pressure is where most weather-related fatal accidents actually get decided, when a pilot overrides a minimum they already knew. *Ref: RMH Ch. 3; PHAK Ch. 2*

D. Cross-Country Flight Planning

56. Walk me through how you planned this cross-country flight. **A:** Selected a route considering airspace, terrain, and available checkpoints; chose a cruising altitude appropriate for terrain, winds, oxygen requirements, and the 91.159 hemispheric rule; obtained a weather briefing and NOTAMs; plotted true course and measured distance; calculated wind correction angle, true heading, groundspeed, and time en route leg-by-leg using winds aloft forecasts; computed fuel burn plus required reserves; verified weight and balance and takeoff/landing performance for the runways to be used; identified alternates/bailout airports along the route. *Ref: AIM 5-1-1; PHAK Ch. 16*

57. What fuel reserve is required for a VFR flight? **A:** Enough to fly to the first point of intended landing and, at normal cruise, continue flying for at least 30 minutes during the day or 45 minutes at night. *Ref: 14 CFR 91.151*

58. Are you required to file a VFR flight plan? **A:** No — voluntary within the US (narrow exceptions like some international/ADIZ crossings). It exists for search-and-rescue protection: Flight Service starts a search if you don't close it within about 30 minutes of your estimated arrival. *Ref: AIM 5-1-4*

59. How do you use a sectional chart to plan a route? **A:** Identify airspace boundaries and floors/ceilings, terrain/obstruction elevations (the maximum elevation figure, MEF, shows the highest obstacle within a quadrant), suitable visual checkpoints, special use airspace, and available nav aids, then plot the true course line with a plotter. *Ref: AIM 9-1; PHAK Ch. 16*

60. What is the difference between true course, magnetic course, magnetic heading, and compass heading? **A:** True course (against true north on the chart) is corrected for magnetic variation to get magnetic course, corrected for wind (wind correction angle) to get true heading, corrected for variation again to get magnetic heading, then corrected for compass deviation (from the compass card) to get compass heading — the number you actually fly. Mnemonic: True Virgins Make Dull Companions (True, Variation, Magnetic, Deviation, Compass); East is Least, West is Best. *Ref: PHAK Ch. 16*

61. What performance/weight-and-balance documents must you check before the flight, and what did you find? **A:** The current, aircraft-specific weight and balance report and equipment list, used with a loading graph to confirm gross weight and CG fall within the approved envelope, plus the takeoff/landing distance charts computed for the day's density altitude, weight, wind, and runway condition. Be ready to show your actual numbers for the airplane and load you're flying that day. *Ref: 14 CFR 91.9; PHAK Ch. 10*

62. What is 14 CFR 91.103, and what must you do before every flight? **A:** Requires the PIC to become familiar with all available information concerning the flight before departure — for a flight not in the vicinity of the departure airport, this includes weather reports/forecasts, fuel requirements, alternatives if the flight can't be completed as planned, and known traffic delays; for any flight, runway lengths and takeoff/landing distance data. *Ref: 14 CFR 91.103*

63. How do you use a winds aloft forecast (FB) in flight planning? **A:** Interpolate the forecast wind direction (true) and speed for your planned altitude and nearest reporting point, then apply it in your groundspeed/heading calculation (E6B or flight computer) for each leg; also check for significant direction/speed change with altitude, which can help you choose a smoother or more favorable cruising altitude. *Ref: AIM 7-1-8; PHAK Ch. 16*

E. National Airspace System

64. What are the six classes of airspace, and how do you identify them on a sectional? **A:** Class A — 18,000 ft MSL to FL600, IFR only, not depicted on VFR sectionals. Class B — around the busiest airports, solid blue rings/shelves, ATC clearance required. Class C — busy radar-served airports, solid magenta rings, two-way radio contact established required. Class D — towered airports without approach control, dashed blue circle, two-way radio contact required. Class E — controlled airspace not otherwise classified, magenta shading/dashed line or blue shading depending on floor. Class G — uncontrolled, not outlined.

Ref: 14 CFR Part 71, 91.129–91.135; AIM 3-2, 3-3

65. What is required to enter Class B, C, and D airspace? **A:** Class B: an explicit ATC clearance, operating two-way radio, Mode C transponder, and ADS-B Out (inside the Mode C veil). Class C and D: establish two-way radio communication with ATC before entering (a simple acknowledgment of your call sign is generally sufficient — not necessarily the word "cleared"); Mode C transponder required in and above Class C. *Ref: 14 CFR 91.129, 91.130, 91.131; AIM 3-2-3 through 3-2-5*

66. What is Special VFR, and what are its rules? **A:** ATC-authorized VFR flight within the surface area of Class B, C, D, or E airspace when weather is below basic VFR minimums but the pilot can remain clear of clouds with at least 1 SM visibility during the day. Must be requested by the pilot (ATC can't require it except in an emergency). At night, both the pilot must be instrument-rated and current and the aircraft IFR-equipped. *Ref: 14 CFR 91.157; AIM 4-4-6*

67. What Special Use and other airspace types should you know, and what do they mean? **A:** Prohibited Area — flight prohibited entirely (e.g., Washington, D.C.). Restricted Area — hazardous activity (artillery/gunnery); entry prohibited when active without permission. Warning Area — same hazards as Restricted but over international waters. MOA — military training; VFR permitted but caution advised. Alert Area — high volume of training/unusual activity; no special clearance required. CFA — activity suspends if a spotter detects an inbound aircraft; not depicted. NSA — pilots requested, not required, to voluntarily avoid. TFR — issued by NOTAM (VIP movement, disasters, sporting events, firefighting); typically mandatory to avoid. *Ref: 14 CFR Part 73, 91.137–91.145; AIM 3-4, 3-5*

68. What are the VFR cruising altitude rules? **A:** Above 3,000 ft AGL in level cruising flight: magnetic course 0°–179° flies odd thousands + 500 ft; magnetic course 180°–359° flies even thousands + 500 ft. *Ref: 14 CFR 91.159*

69. What equipment is required to operate in and around Class B airspace and above 10,000 ft MSL? **A:** A Mode C transponder and ADS-B Out: in Class A, B, and C airspace; in Class E at or above 10,000 ft MSL (excluding at/below 2,500 ft AGL); within 30 NM of a Class B primary airport up to 10,000 ft MSL (the "Mode C veil"); above the ceiling of a Class B or C area up to 10,000 ft MSL. *Ref: 14 CFR 91.215, 91.225*

70. What are the transponder emergency codes? **A:** 1200 — VFR. 7500 — hijacking. 7600 — lost communications. 7700 — general emergency. *Ref: AIM 6-2-2; 14 CFR 91.215*

71. What are the basic right-of-way rules? **A:** An aircraft in distress has right of way over all other traffic. Converging at the same altitude (other than head-on): the aircraft to the other's right has right of way — except a balloon, glider, or towing/refueling aircraft has right of way over powered aircraft. Head-on: both

alter course right. Overtaking: the aircraft being overtaken has right of way; the overtaking aircraft passes to the right. Landing/final approach: the lowest aircraft has right of way, but must not use it to cut off another aircraft already established closer to the runway. *Ref: 14 CFR 91.113*

72. What are NOTAMS, and where do you find them? **A:** Notices to Air Missions — time-critical information not known far enough in advance to be published elsewhere: runway/taxiway closures, NAVAID or lighting outages, temporary obstructions, TFRs, airspace changes. Found via 1800wxbrief.com, ForeFlight, or a Flight Service briefing. *Ref: AIM 5-1-3*

73. What is an ADIZ, and where might you encounter one? **A:** An Air Defense Identification Zone — airspace where an aircraft must file a flight plan, maintain two-way radio contact, and have an operating transponder before entry, for national security identification. The Washington, D.C. SFRA is the one most eastern-US GA pilots need to know. *Ref: 14 CFR 99, 91.161; AIM 3-5-6*

74. What are VFR flyways and VFR corridors, and where would you find them? **A:** A VFR flyway is a charted route (on a VFR Flyway Planning chart, printed on the back of some Terminal Area Charts) helping pilots avoid Class B airspace. A VFR corridor is a defined tunnel through Class B airspace itself, allowing transit without a clearance or ATC contact — both are advisory planning tools, not regulatory shortcuts. *Ref: AIM 3-2-6; Chart User's Guide*

F. Performance and Limitations

75. What atmospheric conditions affect aircraft performance, and how? **A:** Temperature, pressure (altitude), humidity, and wind. Hot/high/humid conditions raise density altitude, reducing air density — cutting lift, propeller/engine efficiency, and thrust — lengthening takeoff/landing rolls and reducing climb performance. Headwind shortens ground roll/distance; tailwind lengthens both significantly. *Ref: PHAK Ch. 10, 11*

76. What is pressure altitude, and what is density altitude? **A:** Pressure altitude is the altitude indicated when the altimeter is set to 29.92 inHg. Density altitude is pressure altitude corrected for nonstandard temperature — the altitude the aircraft/engine "feel" they're performing at, and the number performance charts are entered with. *Ref: PHAK Ch. 4, 11*

77. How does weight affect performance? **A:** Increasing gross weight increases stall speed, takeoff/landing distance, and required runway length; decreases climb rate, service ceiling, and maneuverability; reduces the margin within the structural envelope. *Ref: PHAK Ch. 10*

78. How does CG location (forward vs. aft) affect performance and stability? **A:** Forward CG: increases longitudinal stability, slightly increases stall speed, requires more nose-up trim/elevator authority (especially in the landing flare), generally improves stall/spin resistance, slightly reduces cruise speed. Aft CG: decreases longitudinal stability, slightly decreases stall speed, reduces the elevator/control margin for stall recovery, makes stalls and spins more difficult (potentially unrecoverable) — the more dangerous condition. *Ref: PHAK Ch. 10; AFH Ch. 4*

79. What are the four forces acting on an airplane, and how do they relate in each flight regime? **A:** Lift, weight, thrust, and drag. In steady, unaccelerated, straight-and-level flight, lift equals weight and thrust

equals drag. In a climb, thrust must exceed drag by the amount needed to overcome the added weight component; in a descent, less thrust than drag is needed. *Ref: PHAK Ch. 5*

80. What risks come from exceeding an airplane's limitations? **A:** Exceeding max gross weight or the CG envelope degrades performance/controllability and can prevent stall/spin recovery. Exceeding V_a with full/abrupt control inputs, or load-factor limits in turbulence/steep maneuvering, can cause structural damage. Exceeding max operating altitude or V_{ne} risks structural/control-surface damage. All limitations are found in the POH limitations section — a legal document. *Ref: 14 CFR 91.9; PHAK Ch. 10*

Aerodynamics Fundamentals

81. What is angle of attack? **A:** The acute angle between the wing's chord line (leading edge to trailing edge) and the relative wind — the direction the air is actually meeting the wing. *Ref: PHAK Ch. 5*

82. When does an aircraft stall, and at what airspeed? **A:** A stall occurs whenever the wing exceeds its critical angle of attack, causing airflow over the upper surface to separate and lift to decay rapidly — it can happen at any airspeed, pitch attitude, or power setting. Indicated stall speed only describes the specific 1G, coordinated, given-weight-and-configuration condition most often demonstrated. *Ref: PHAK Ch. 5; AFH Ch. 4*

83. How does a wing produce lift? **A:** Two complementary principles. Bernoulli's principle: as air speeds up over the wing's curved upper surface, static pressure drops, creating a pressure differential. Newton's third law: the wing deflects air downward and rearward; the equal-and-opposite reaction pushes the wing up. Both act together. *Ref: PHAK Ch. 5*

84. What is load factor? **A:** The ratio of the load the wings are producing (lift) to the aircraft's actual weight, in G's — 2G means the wings produce lift equal to twice the airplane's weight. Load factor increases with bank angle in a level turn (2G at 60° bank) and increases stall speed by the square root of the load factor. *Ref: PHAK Ch. 5*

85. What are the four left-turning tendencies, and when is each most pronounced? **A:** Torque reaction — Newton's third law; the engine spins the prop right, so an equal/opposite reaction rolls the airframe left; most pronounced at low airspeed/high power. P-factor — asymmetric propeller loading; at high AOA the descending blade takes a bigger "bite," yawing the nose left; most pronounced at high power, low airspeed, high AOA. Spiraling slipstream — corkscrewing propwash strikes the left side of the tail, yawing the nose left; most pronounced at low airspeed/high power. Gyroscopic precession — a force on a spinning gyroscope is felt 90° later in the direction of rotation; most pronounced when the propeller's plane of rotation is displaced. *Ref: PHAK Ch. 5; AFH Ch. 3*

86. What is load factor's relationship to stall speed, and what is maneuvering speed (V_a)? **A:** Stall speed increases with the square root of load factor (a 2G load increases it ~41%). V_a is the maximum speed at which a single, full, abrupt control deflection will stall the wing before exceeding the airplane's structural limit — it protects the airframe by ensuring the wing stalls (shedding load) before the structure fails. V_a decreases as weight decreases, since a lighter airplane stalls sooner. *Ref: PHAK Ch. 5; AFH Ch. 4*

87. What is ground effect? **A:** A reduction in induced drag (and slight increase in effective lift) when the wing operates within roughly one wingspan of the ground, since the ground restricts wingtip vortex formation. It can cause premature liftoff before best-rate-of-climb speed, and float during landing if excess airspeed carries into the flare. *Ref: PHAK Ch. 5*
88. What is the difference between indicated, calibrated, true, and groundspeed? **A:** Indicated (IAS) is what the instrument reads. Calibrated (CAS) corrects IAS for instrument/position error. True (TAS) corrects CAS for nonstandard pressure/temperature (density altitude) — increases relative to CAS with altitude. Groundspeed is TAS corrected for wind. *Ref: PHAK Ch. 4*
89. What is adverse yaw, and how do ailerons/rudder counteract it? **A:** Rolling into a turn, the down-going aileron on the rising wing produces more lift and induced drag than the up-going aileron on the lowering wing, yawing the nose toward the outside of the turn. Coordinated rudder in the direction of the turn counteracts it. *Ref: PHAK Ch. 5*

G. Operation of Systems

90. What type of engine does your aircraft have? **A:** A Continental O-300-C: six-cylinder, horizontally opposed, air-cooled, direct-drive, normally aspirated, carbureted, rated 145 BHP at 2,700 RPM, driving a fixed-pitch McCauley propeller (76 in.). Note: every O-300 variant is six-cylinder — don't confuse it with the 4-cylinder Lycoming O-320/O-360 found in most later-model 172s. *Ref: 1965 Cessna 172/Skyhawk Owner's Manual §3-3; PHAK Ch. 7*
91. How does the ignition/magneto system work, and why are there two magnetos? **A:** Two spark plugs per cylinder, each fired by an independent, engine-driven magneto needing no aircraft electrical power — ignition continues through a total electrical failure. On this aircraft, the magneto check is at 1,600 RPM; the difference in RPM drop between the two magnetos checked individually must not exceed 75 RPM. No RPM drop at all when switching to a single magneto is itself a red flag (possible grounding fault or advanced timing), not a "good" result. *Ref: Owner's Manual §2-6; N8821U checklist*
92. Describe the fuel system. **A:** Gravity-fed — no engine-driven or electric fuel pump. Fuel flows from two wing tanks through a selector valve (OFF/LEFT/RIGHT/BOTH) and a strainer to the carburetor. 19.5 gal/tank total (39 gal total); 18.0 gal/tank usable in all conditions, up to 19.0 gal/tank in level flight only. Standard procedure is takeoff/climb/landing on BOTH; the manual permits switching to a single tank in cruise. Minimum fuel grade per this POH is 80/87; N8821U runs 100LL today, an approved higher-octane substitute. *Ref: Owner's Manual §2-1, 2-2, Fig. 2-1/2-2*
93. What causes carburetor ice, and how do you detect and correct it? **A:** Can form even in outside air temperatures well above freezing (roughly 20°F–70°F with visible moisture or high humidity), from the venturi and fuel-vaporization cooling effect in the carburetor throat. On a fixed-pitch prop, the first symptom is an unexplained, gradual RPM loss and roughness. Correction: apply full carb heat; expect a further momentary RPM drop as the ice melts, followed by a rise as it clears. The POH cautions against partial carb heat in sub-zero conditions, since it can bring carb air temperature into the 32–80°F range where icing is actually most critical. *Ref: Owner's Manual §2-4; PHAK Ch. 7*

94. Describe the electrical system. **A:** A 14-volt DC system powered by an engine-driven alternator — N8821U was converted from the original factory generator (alternators produce usable current at lower RPM, e.g. idle). A 12-volt battery on the left forward firewall provides starting/backup power. All circuits except the clock are on push-to-reset breakers, except the stall warning horn/turn coordinator circuit, which shares a single automatically-resetting breaker per the original type design. *Ref: Owner's Manual §2-3*
95. What are your aircraft's primary and secondary flight controls? **A:** Primary: ailerons (roll), elevator (pitch), rudder (yaw). Secondary: flaps and trim. N8821U's trim is a single elevator/pitch trim tab; there is no separate rudder trim. *Ref: PHAK Ch. 5; Owner's Manual §1-2*
96. What landing gear and hydraulic systems does your aircraft have? **A:** Fixed tricycle gear, with an oleo-pneumatic nose strut and torque links, serviced with MIL-H-5606 hydraulic fluid — the same fluid used in the toe-brake system. *Ref: Owner's Manual §4-7, inside back cover*
97. Does your aircraft have any deicing/anti-icing equipment? **A:** Nothing is listed in this POH — no pitot heat is called out. Treat the aircraft as having no anti-icing equipment absent a specific placard/logbook entry, and it is not approved for flight into known icing regardless. *Ref: Owner's Manual, Section VI*
98. What optional systems does N8821U's POH describe, and would you expect them installed? **A:** An optional auxiliary fuel tank (18-gallon baggage-compartment tank with electric transfer pump), optional landing lights, rotating beacon, and a Nav-O-Matic autopilot. Since N8821U's checklist doesn't reference a fuel transfer pump, treat the standard two-tank gravity system as installed unless confirmed otherwise. *Ref: Owner's Manual Section VI*

H. Human Factors

99. What is hypoxia? Give the types, symptoms, and corrective action. **A:** Insufficient oxygen to body tissue, particularly the brain. Four types: hypoxic (altitude), hypemic (blood can't carry O₂ — anemia, CO poisoning), stagnant (poor circulation — cold, high G), histotoxic (cells can't use O₂ — alcohol/drugs). Symptoms: euphoria, poor judgment, headache, drowsiness, tunnel vision, bluish lips/nail beds, eventual unconsciousness — often unrecognized by the pilot. Corrective action: descend, use supplemental oxygen, treat the cause. *Ref: PHAK Ch. 17; AIM 8-1-2*
100. At what altitudes is supplemental oxygen required? **A:** Required crew must use it above 12,500 ft MSL through 14,000 ft MSL for the portion of flight lasting more than 30 minutes, and continuously above 14,000 ft MSL. Above 15,000 ft MSL, every occupant must be provided (not just offered) supplemental oxygen. *Ref: 14 CFR 91.211*
101. What is hyperventilation? Symptoms and correction? **A:** Abnormally rapid/deep breathing that washes out CO₂ faster than the body produces it. Symptoms mimic hypoxia: dizziness, tingling extremities, hot/cold sensations, possible unconsciousness. Usually triggered by stress/anxiety/fear. Correction: consciously slow breathing, talk aloud, treat the underlying stressor. (Avoid the old "paper bag" fix — at cruise altitude, rebreathing exhaled air can tip someone from hyperventilating straight into hypoxia.) *Ref: PHAK Ch. 17*
102. What causes ear and sinus problems in flight, and how do you prevent/correct them? **A:** Pain from the eustachian tubes being unable to equalize pressure between the middle ear/sinuses and cabin, usually

during descent, worsened by congestion. Correct with the Valsalva maneuver (pinch nose, gently blow), yawning, or swallowing; best prevention is not flying with an upper respiratory infection or bad allergies. *Ref: PHAK Ch. 17*

103. What is spatial disorientation, and how is it avoided? **A:** A false perception of the aircraft's position/attitude/motion from conflicting or absent visual, vestibular, and proprioceptive cues — most likely at night or in IMC. Avoided by trusting the flight instruments over physical sensations, maintaining instrument proficiency, and avoiding flight beyond your training/currency. *Ref: PHAK Ch. 17*

104. What is carbon monoxide poisoning, and how do you guard against it? **A:** A colorless, odorless gas from engine exhaust that can leak into the cabin (commonly through a cracked heat exchanger feeding the cabin heater) and binds to hemoglobin far more readily than oxygen, causing hypemic hypoxia. Symptoms (headache, dizziness, confusion, drowsiness) mimic fatigue. Response: cabin heat off, fresh air vents open, land as soon as practical; carry and heed a CO detector. *Ref: PHAK Ch. 17*

105. What optical illusions should a pilot be aware of? **A:** Runway width/slope illusions (narrower runway feels too high, wider too low; upslope feels too high, downslope too low), featureless terrain illusion (causes a lower-than-normal approach), black hole approach (dark terrain/water makes the runway seem farther/higher), autokinesis (a stationary light at night appears to move after being stared at). Countered by cross-checking instruments and glidepath references (VASI/PAPI) over visual impression. *Ref: PHAK Ch. 17*

106. How long should you wait to fly after scuba diving? **A:** At least 12 hours after a no-decompression dive; at least 24 hours after any dive requiring a decompression stop, or after multiple/repetitive dives. Flying too soon risks decompression sickness as reduced cabin pressure lets dissolved nitrogen form bubbles in the bloodstream. *Ref: AIM 8-1-2*

107. What is Aeronautical Decision-Making (ADM), and what are SRM/CRM? **A:** ADM is a systematic, continuous process for identifying hazards, assessing risk, and choosing the best course of action given available resources and time. CRM (Crew Resource Management) is the use of all available resources by a multi-pilot crew; SRM (Single-Pilot Resource Management) is the same concept applied by one pilot managing all internal and external resources alone. *Ref: PHAK Ch. 2; RMH Ch. 1*

108. What are the five hazardous attitudes, and what is the antidote to each? **A:** Anti-authority ("don't tell me") — follow the rules, they're usually right. Impulsivity ("do it now") — not so fast, think first. Invulnerability ("it won't happen to me") — it could happen to me. Macho ("I can do it") — taking chances is foolish. Resignation ("what's the use") — I'm not helpless, I can make a difference. *Ref: PHAK Ch. 2*

109. How do you manage distractions, task prioritization, and loss of situational awareness? **A:** Prioritize using "aviate, navigate, communicate," follow sterile-cockpit discipline during critical phases (taxi, takeoff, approach, landing), use checklists rather than memory, and maintain a continuous instrument/outside cross-check to avoid fixating on any one task. *Ref: PHAK Ch. 2, 17*

110. What's the difference between knowing your personal minimums and actually living by them? **A:** Knowing them is just a number on paper. Living by them means holding to that number under real pressure — a passenger's schedule, a bill that needs the flight, "get-there-itis" — which is where most weather-related fatal accidents actually originate: not from ignorance of the risk, but from a pilot overriding a minimum they already knew. *Ref: PHAK Ch. 2; RMH Ch. 3*

II. Preflight Procedures

A. Preflight Assessment

111. How do you conduct a preflight inspection, and what are you really looking for? **A:** Follow the POH checklist systematically (not from memory), assessing structure, control surfaces, fuel quantity/quality (visual check plus sump drains), oil quantity, tire condition, brakes, static ports/pitot, lights, propeller condition, and cowlings/inspection panel security — while also assessing weather, NOTAMs, and performance/loading. The goal isn't just touching every item, it's forming a go/no-go judgment. *Ref: AFH Ch. 2; POH Section 4*

112. What would you do if you found a discrepancy during preflight? **A:** Determine whether the item is required equipment (91.205/MEL/KOEL/AD). If so, the aircraft is unairworthy until repaired or properly deferred by a mechanic. If not, it can potentially be placarded inoperative under 91.213(d). Either way, notify a mechanic and log it — the PIC has final authority and responsibility for the airworthiness decision. *Ref: 14 CFR 91.7, 91.213*

B. Flight Deck Management

113. What should a passenger safety briefing include? **A:** Seatbelts/shoulder harnesses, seat position and door operation, checklist use, sterile cockpit expectations, location/use of emergency equipment, smoking policy, and headset/intercom use. For pre-solo or new passengers, also cover what to do in an emergency. *Ref: AIM 4-1-1; AFH Ch. 2*

114. How do you manage checklists, and what's the difference between a "do-list" and a "flow"? **A:** Checklists (or a memorized flow verified against a checklist) ensure critical items aren't missed, especially before takeoff and landing. A "do-list" is read and performed item by item in real time (less time-critical phases); a memorized "flow" is performed from memory and then verified against the printed checklist, commonly used for time-critical phases so attention stays outside. *Ref: AFH Ch. 2*

C. Engine Starting

115. Describe a normal engine start, and what would indicate a problem? **A:** Confirm the area is clear, set the parking brake, follow the POH starting checklist (mixture rich, throttle cracked, master/fuel pump on as required, prime as needed, "clear prop," engage starter), and monitor oil pressure — it should rise to normal range within about 30 seconds; if not, shut down. A start that won't catch after several attempts should follow the POH's abnormal-start procedure rather than repeated cranking, which risks starter damage. *Ref: POH Section 4; AFH Ch. 2*

116. What would you do if a fire occurred during engine start? **A:** Continue cranking (to draw the fire into the intake/cylinders and extinguish it) while calling for mixture to idle cutoff and fuel selector off; if the fire

doesn't clear, shut down the master and evacuate, using an extinguisher on an external fire if available. *Ref: AFH Ch. 2; POH Section 3*

117. What would you do for a hot start or a flooded engine on a carbureted aircraft like this one? **A:** A hot start often needs little or no priming, sometimes leaning slightly during cranking. A flooded (over-primed) engine is typically cleared with the checklist's flooded-start procedure — often throttle open, mixture idle cutoff, crank until it catches, then bring mixture to rich — rather than repeated normal-start attempts, which just adds more fuel. *Ref: POH Section 4; AFH Ch. 2*

D. Taxiing

118. How do you prevent a runway incursion? **A:** Know and follow airport signage/markings (a red-background sign is a mandatory instruction), always stop and hold short at a solid double-yellow hold line unless cleared onto/across the runway, read back all runway crossing/hold-short clearances, and stop and ask if in doubt about your position or a clearance rather than proceeding. *Ref: AIM 4-3-18, 4-3-19; AC 91-73*

119. How do you position the flight controls while taxiing in wind? **A:** "Climb away from a tailwind, dive away from a headwind": quartering headwind — aileron into the wind, elevator neutral-to-forward; quartering tailwind — aileron away from the wind (down on the tailwind side), elevator forward — keeping wind from getting under the wing or tail. *Ref: AFH Ch. 2, 13*

120. What should you do if the brakes fail while taxiing? **A:** Reduce power to idle, use the parking brake if independent, and if stopping distance is insufficient, steer toward open/soft ground to slow rather than risk collision; shut down (mixture idle cutoff) if a stop can't otherwise be assured. *Ref: AFH Ch. 2*

F. Before Takeoff Check

121. What do you check/verify during the runup and before-takeoff check? **A:** Following the POH checklist: parking brake set, engine temps/pressures in the green, magneto check (expected RPM drop within limits, no dead-cutting), carb heat check, flight instruments set, flight controls free and correct, flaps set for takeoff, trim set, doors/seatbelts secure, and a review of the takeoff briefing (abort points, emergency actions) and performance numbers. *Ref: POH Section 4; AFH Ch. 2, 5*

122. What would you do if you found an unacceptable magneto RPM drop or no drop at all during runup? **A:** An excessive drop usually means a fouled plug, bad wire, or timing issue — try leaning aggressively and cycling power once to clear a fouled plug, then taxi back if it doesn't clear; don't depart. No drop at all (the "hot mag" case) suggests a grounding/switch fault — also not airworthy, since it can mean the engine can't be shut off with the ignition switch. *Ref: POH Section 4; AFH Ch. 2*

III. Airport and Traffic Pattern Operations

A. Communications, Light Signals, and Runway Lighting Systems

123. How do you determine the correct frequency to use, and what is standard radio phraseology? **A:** Use the sectional chart or chart supplement/airport diagram for CTAF, tower, ground, and ATIS/AWOS/ASOS frequencies; state aircraft identification, position, altitude, and intentions concisely using standard phraseology. *Ref: AIM 4-2, 4-1-9; Chart Supplement*

124. What are the ATC light gun signals? **A:** Steady green — cleared for takeoff (ground) / cleared to land (flight). Flashing green — cleared to taxi / return for landing. Steady red — stop / give way, continue circling. Flashing red — taxi clear of runway / airport unsafe, do not land. Alternating red/green — exercise extreme caution in either case. *Ref: AIM 4-3-13*

125. What is lost communication procedure at a towered airport? **A:** Squawk 7600, remain outside or above Class D/C/B airspace if unsure of clearance, watch for light gun signals; if arriving, join the pattern and look for light signals. Continue to comply with the last clearance received. *Ref: AIM 4-2-13; 14 CFR 91.185*

126. Describe runway lighting and marking systems you might encounter. **A:** Runway edge lights white (yellow for last 2,000 ft on instrument runways); threshold lights green (approach) / red (from the runway); REILs are synchronized flashing white lights marking the threshold. VASI (2-bar): red over white = on glidepath ("red over white, you're all right"), white over white = too high, red over red = too low. PAPI (4 lights): 2 red/2 white = on glidepath, more white = high, more red = low. *Ref: AIM 2-1, 2-3*

127. What happens to a Class D airport's airspace when the tower closes for the night? **A:** Typically reverts to Class E (if it has surface-based weather reporting) or Class G, per times published in the chart supplement — no ATC communication required while closed; normal nontowered self-announce procedures apply. *Ref: AIM 3-2-5; Chart Supplement*

B. Traffic Patterns

128. Describe a standard traffic pattern. **A:** A rectangular pattern flown to the left (unless otherwise indicated) at pattern altitude (commonly 1,000 ft AGL for a piston single — check the chart supplement), consisting of upwind, crosswind, downwind, base, and final legs, maintaining appropriate spacing/altitude/airspeed while watching for traffic and using appropriate radio calls at a nontowered field. *Ref: AIM 4-3-2, 4-3-3; AC 90-66*

129. How do you determine which runway/pattern direction to use at a nontowered airport, and what's the recommended pattern entry? **A:** Check the segmented circle and wind sock/tee for the active runway and pattern direction, and the chart supplement for standard pattern altitude/direction. Recommended entry: 45° to the downwind leg, at pattern altitude, entering on the midpoint of the downwind. *Ref: AIM 4-3-2, 4-3-3*

130. What are the standard right-of-way and communication procedures at a nontowered field? **A:** Self-announce position and intentions on CTAF (no clearance is issued — pilots see-and-avoid and self-sequence); follow standard right-of-way rules; an aircraft on final has priority, but must not use it to cut off another aircraft already established closer to the runway. *Ref: AIM 4-1-9; 14 CFR 91.113, 91.126*

131. Two aircraft with no radio contact converge to land at the same nontowered airport. Who has right of way? **A:** The aircraft at the lower altitude has right of way, but may not use that right to cut off another aircraft on final approach or to overtake it. An aircraft already established on final generally has priority regardless of how it got there. *Ref: 14 CFR 91.113(g); AIM 4-3-3*

IV. Takeoffs, Landings, and Go-Arounds — Knowledge Review

(The skill portions of this Area are demonstrated in flight; the oral typically probes the knowledge below.)

132. What are the differences between a normal, short-field, and soft-field takeoff? **A:** Normal — flaps per POH, smooth acceleration, rotate at POH speed. Short-field — max performance, brakes held with full power before release, rotate at V_x to clear an obstacle, then accelerate to V_y . Soft-field — full aft yoke to unload the nosewheel, lift off in ground effect, accelerate to climb speed before climbing out. On N8821U: flaps up (0°) for both normal and short-field/obstacle takeoffs — 10° flaps shortens the ground roll but that's lost by the 50-ft obstacle point per POH. Rotate at 60 MPH; V_x 65 MPH; V_y 80 MPH. *Ref: Owner's Manual §2-7; N8821U checklist*

133. What are V_x and V_y , and how do they change with altitude? **A:** V_x = best angle of climb (max altitude per horizontal distance) — obstacle clearance. V_y = best rate of climb (max altitude per time) — normal climb. V_x increases and V_y decreases with altitude until they converge at absolute ceiling. On N8821U: V_y 80 MPH (sea level) to 77 MPH (10,000 ft); V_x 65 MPH (sea level) to 71 MPH (10,000 ft). *Ref: PHAK Ch. 11; Owner's Manual §2-8, §5-3*

134. What is the decision-making process for a go-around, and when should you commit to one? **A:** Go around any time the approach isn't stabilized, the runway isn't clear, or anything looks wrong — decide early rather than trying to salvage a bad approach. On N8821U: full power, flaps to 20° immediately, then retract slowly to fully up only after reaching a safe airspeed. *Ref: Owner's Manual §2-8; AFH Ch. 8*

135. What is the "impossible turn"? **A:** Turning back to the departure runway after an engine failure on takeoff — usually not enough altitude/energy to complete without stalling. Brief it before every takeoff: plan to land essentially straight ahead ($\pm 30^\circ$) below a pre-briefed altitude, and only turn back if practiced/briefed and above that altitude. On N8821U, best glide is V_g 80 MPH — establish it immediately either way. *Ref: AFH Ch. 5, 16; N8821U checklist*

V. Performance and Ground Reference Maneuvers — Knowledge Review

136. What is the knowledge tested behind steep turns? **A:** Load factor and stall speed increase with bank angle (2G/41% increase at 60°), requiring added back-pressure/power to hold altitude, and overbanking tendency requiring opposite aileron. ACS standard: altitude ± 100 ft, bank $\pm 5^\circ$, rollout heading $\pm 10^\circ$. On N8821U, POH steep-turn entry speed is 122 MPH (106 knots) — same as V_a . *Ref: PHAK Ch. 5; AFH Ch. 9; Owner's Manual §3-2; ACS Area V.A*

137. What is wind drift correction, and why is it needed on ground reference maneuvers (turns around a point, S-turns, rectangular course)? **A:** Bank angle must vary to hold a constant radius/track over the ground despite wind — steepest downwind (highest groundspeed), shallowest into the wind (lowest groundspeed). Builds the wind-correction judgment needed for the pattern and precision approaches. *Ref: AFH Ch. 6*

VI. Navigation

A. Pilotage and Dead Reckoning

138. What is the difference between pilotage and dead reckoning? **A:** Pilotage is navigating by visually matching ground landmarks to the chart. Dead reckoning is navigating by calculation — heading, groundspeed, and time from true course, wind correction, and true airspeed — without reference to landmarks. VFR cross-country normally combines both. *Ref: PHAK Ch. 16*

139. What magnetic compass errors should you understand? **A:** Variation — the fixed angular difference between true and magnetic north at a location. Deviation — compass card error from the aircraft's own magnetic/electrical fields, unique to each aircraft (posted on its correction card). Dip errors during maneuvering: north/south headings, the compass leads/lags in turns ("ANDS" — Accelerate North, Decelerate South); east/west headings, acceleration/deceleration alone shows a turn that isn't happening. *Ref: PHAK Ch. 8*

140. What is your process for selecting checkpoints, route, and altitude? **A:** Checkpoints: prominent, unambiguous features spaced roughly 10–15 minutes apart. Route: balances directness against airspace, terrain/obstacles, and emergency landing options. Altitude: terrain/obstacle clearance, the 91.159 hemispheric rule, winds aloft, and oxygen/comfort for longer legs. *Ref: PHAK Ch. 16*

141. What is the difference between groundspeed and true airspeed, and how does wind affect them? **A:** True airspeed is the aircraft's speed through the air mass; groundspeed is speed over the ground, equal to true airspeed adjusted for wind (headwind reduces it, tailwind increases it, crosswind requires a wind correction angle). *Ref: PHAK Ch. 16*

B. Navigation Systems and Radar Services

142. How does VOR navigation work? **A:** A VOR ground station lets an aircraft receiver determine its magnetic bearing (radial) from the station. Tune and identify the station, set the desired course on the OBS, and read the CDI — centered with a "TO" or "FROM" flag shows you're on that radial's course; each dot of deflection is about 2° off course. *Ref: PHAK Ch. 16; AIM 1-1-3*

143. How do you check that a VOR receiver is functioning correctly, and how often is it required? **A:** Using a VOT test signal, a certified checkpoint, or comparing two receivers against each other — required only if the VOR will be used for IFR navigation, logged within the preceding 30 days. *Ref: 14 CFR 91.171*

144. What are the advantages and limitations of GPS versus VOR? **A:** GPS gives continuous, accurate position, ground track, and direct point-to-point navigation independent of ground infrastructure, but depends on satellite signal integrity and database currency. VOR is a simpler, independent, ground-based backup that doesn't rely on GPS, but only defines specific radials from fixed stations and needs more workload for direct navigation. *Ref: PHAK Ch. 16*
145. What radar services are available to VFR pilots? **A:** VFR flight following (traffic advisories, workload-permitting) plus radar-based lost-procedures assistance, vectors around weather, and Class B/C traffic sequencing. *Ref: AIM 4-1-15, 4-1-18*
146. What is RAIM, and why does it matter for GPS navigation? **A:** Receiver Autonomous Integrity Monitoring — a GPS receiver's internal check that enough satellites are in view and well-positioned to detect a faulty signal. A RAIM warning means the GPS position can't be trusted and you should cross-check with other means. *Ref: AIM 1-1-17*

C. Diversion

147. How do you divert to an alternate airport in flight? **A:** Identify a suitable alternate on the chart (fuel, runway suitability, weather); turn toward it promptly; estimate heading and distance using the chart or a simple rule of thumb rather than exact math while flying; update ATC or Flight Following if in contact. *Ref: PHAK Ch. 16*
148. What should you do if you're unsure you have enough fuel to reach your destination? **A:** Divert early to the nearest suitable airport while there's still a comfortable margin — don't wait until it's critical. Recalculate conservatively, and land and refuel if there's genuine doubt; fuel exhaustion is one of the most preventable causes of GA accidents. *Ref: 14 CFR 91.151; AIM 6-2-1*

D. Lost Procedures

149. What do you do if you become lost or disoriented in flight? **A:** The "5 C's": Climb (better visual range/reception), Communicate (contact ATC or Flight Service — 121.5 MHz is the emergency frequency), Confess you're unsure of position, Comply with instructions given, Conserve fuel (best-range settings). Also maintain heading and look for large, distinctive landmarks to correlate with the chart. *Ref: PHAK Ch. 16; AIM 6-2-1*

VII. Slow Flight and Stalls

A. Maneuvering During Slow Flight

150. What is the purpose of practicing slow flight, and what changes about the airplane's handling? **A:** Builds proficiency to control the airplane, recognize stall onset (buffet, horn, reduced control effectiveness),

and safely maneuver near stall speed (pattern, takeoff, landing). At high AOA/low airspeed: less control effectiveness, more adverse yaw, stronger P-factor, smaller stall margin. *Ref: AFH Ch. 4; ACS Area VII.A*

B/C. Power-Off and Power-On Stalls

151. What are the differences between power-off and power-on stalls, and what do they simulate? **A:** Power-off simulates an approach-to-landing stall (landing configuration, idle power, gradual pitch-up) — the more common real-world accident scenario. Power-on simulates a takeoff/departure stall (takeoff configuration, climb power) — a more abrupt break with stronger P-factor yaw. *Ref: AFH Ch. 4*

152. What is the proper stall recovery procedure? **A:** At the first indication: reduce AOA (release back-pressure), add max available power smoothly, level the wings with coordinated aileron/rudder, minimize altitude loss. Don't try to pick up a dropped wing with aileron alone — use rudder. *Ref: AFH Ch. 4*

153. What is a secondary stall, and what is an accelerated stall? **A:** Secondary — pitching up too abruptly during recovery, re-exceeding critical AOA before regaining airspeed. Accelerated — a stall at higher-than-normal airspeed from increased load factor (e.g., a steep turn) — critical AOA, not a specific airspeed, always triggers a stall. *Ref: AFH Ch. 4*

154. Why is a cross-control stall especially dangerous? **A:** It's typically uncoordinated — rudder and aileron working against each other while overshooting the turn from base to final — causing one wing to stall before the other. That asymmetric stall can snap the aircraft into a spin at pattern altitude, too low to recover. *Ref: AFH Ch. 4*

D. Spin Awareness

155. What causes a spin, and what is the recovery procedure? **A:** A spin is an aggravated stall with one wing more stalled than the other, causing autorotation — requires a stall plus a yawing input (uncoordinated flight). Standard recovery (PARE, generic — always follow your POH): Power idle, Ailerons neutral, Rudder full opposite the rotation, Elevator briskly forward to break the stall, then neutralize and recover from the dive. *Ref: AFH Ch. 4; PHAK Ch. 5*

156. What conditions most commonly lead to an unintentional spin? **A:** Uncoordinated flight at high AOA/low airspeed — classically, skidding the base-to-final turn to avoid overshooting centerline, or an uncoordinated climbing turn after takeoff. It has to be immediate and reflexive, since this is exactly where an inadvertent spin most often happens: too low for a checklist. *Ref: AFH Ch. 4*

157. What causes a spin, and how does it differ from a spiral dive? **A:** A spin always begins with a stall — an uncoordinated stall where one wing is stalled more than the other, causing autorotation around a mostly-vertical axis internal to the aircraft. A spiral is not stalled at all — a steepening diving turn rotating around an external point, with airspeed increasing rather than staying near stall speed. *Ref: AFH Ch. 4; PHAK Ch. 5*

VIII. Basic Instrument Maneuvers

158. Why does a private pilot need any instrument flying skill? **A:** Per 61.109's 3-hour instrument training requirement, a private pilot must be able to control the aircraft solely by instruments long enough to escape inadvertent IMC — a survival skill, not an instrument-rating substitute. *Ref: 14 CFR 61.109(a)(3); ACS Area VIII*

159. What is the basic instrument scan, and what's the "primary/supporting" method? **A:** A continuous cross-check of the attitude indicator, airspeed, altimeter, VSI, heading indicator, and turn coordinator, rather than fixating on one. Each instrument is primary for one parameter (pitch, bank, or power) and supports interpretation of the others. *Ref: IFH Ch. 3, 4*

160. What are common unusual attitudes, and how do you recover? **A:** Nose-low/increasing airspeed: reduce power, level the wings, smoothly raise the nose. Nose-high/decreasing airspeed: add full power, level the wings, smoothly lower the nose. Correct bank and pitch together, referencing the attitude indicator, and avoid abrupt inputs. *Ref: IFH Ch. 3; AFH Ch. 4*

161. If you enter spatial disorientation, what's the first instrument you should reference to begin recovery — and why not the attitude indicator? **A:** The airspeed indicator. The attitude indicator is a gyroscopic instrument that can tumble and give an unreliable reading once bank exceeds roughly 60° on older vacuum instruments — exactly the situation you're in during disorientation. Airspeed trending fast means the nose is likely low; airspeed trending toward a stall warning means the nose is likely high. Stabilize on airspeed first, then cross-check the rest. *Ref: IFH Ch. 3; AFH Ch. 4*

162. What should you do if you inadvertently enter IMC? **A:** Transition to instruments, hold wings level and constant airspeed, avoid abrupt inputs, and turn back toward known VFR conditions (standard-rate 180°, roughly bank angle equal to the last two digits of airspeed in knots). If unable, contact ATC, advise you're a non-instrument-rated pilot inadvertently in IMC, and declare an emergency (squawk 7700) if needed. *Ref: AIM 8-1-3; AFH Ch. 4*

IX. Emergency Operations

A. Emergency Descent

163. When and how would you perform an emergency descent? **A:** Used to descend as rapidly as possible within structural limits — escaping smoke/fire, a pressurization problem, or severe icing above a warmer layer. Reduce power to idle, establish the POH-specified configuration/airspeed, and bank moderately to maintain visual scan and control. N8821U's checklist: throttle idle, descending turns at 30°–45° bank, airspeed 120 MPH, declare an emergency (120 MPH is comfortably below V_{ne} of 174 MPH, inside the yellow caution arc — smooth air only). *Ref: POH Section 3; AFH Ch. 16; N8821U checklist*

B. Emergency Approach and Landing (Simulated)

164. What are your immediate actions after a complete engine failure in flight? **A:** Fly the airplane first: pitch immediately for best glide speed. Pick the best available landing site within gliding distance (wind, surface, obstacles, length). Run the engine-restart checklist while continuing toward the chosen field. If restart fails, secure the engine if time allows, communicate (7700, radio call) if time/altitude allow, and brief passengers before configuring for landing. N8821U's checklist: airspeed Vg 80 MPH, carb heat ON, fuel selector BOTH, flaps UP, mixture RICH, primer IN AND LOCKED, magnetos check ALL. If the prop isn't windmilling, ignition to START and check magnetos again, then declare an emergency and secure the engine if restart fails. Once committed to landing: select/inspect the landing area, maintain Vg 80 MPH, mixture CUTOFF, fuel selector OFF, ignition/electrical OFF, flaps as required, master OFF, doors unlatched prior to touchdown, brakes applied heavily after touchdown. *Ref: POH Section 3; AFH Ch. 16; N8821U checklist*

165. Engine fails at 200 feet right after takeoff. What do you do? **A:** Land straight ahead (minor turns only to avoid an obstacle), pitch immediately for best glide. At that altitude there is essentially never enough height to turn back — the "impossible turn" is one of the classic fatal-accident scenarios in GA. Brief the plan before every takeoff. *Ref: AFH Ch. 5, 16*

166. How do you choose a forced-landing site, and what pattern do you fly to it? **A:** Prioritize a site that's long, into the wind if practical, free of obstacles, and reachable without maneuvering below pattern altitude too early — a firm open field beats stretching the glide to an airport. Fly a modified pattern (a "key position" abeam the intended touchdown point) to manage energy, using flaps/slips to adjust glidepath, and plan to touch down at minimum controllable airspeed. Remember: N8821U's POH prohibits slips with full (40°) flaps — favor a lesser flap setting if a slip will be needed. *Ref: AFH Ch. 16; Owner's Manual §2-9*

C. Systems and Equipment Malfunctions

167. What would you do for a partial power loss (rough-running engine)? **A:** Apply carburetor heat first (the most common cause of partial power loss is carb ice), check mixture, check magnetos individually (a fouled plug may run better on one), check fuel selector/quantity and switch tanks if warranted, and scan engine gauges for a diagnosable cause. If power is sufficient, divert to the nearest suitable airport; if it keeps degrading, treat it as an impending complete failure. *Ref: AFH Ch. 16; POH Section 3*

168. What would you do for an electrical system failure? **A:** Recognize it — on N8821U, a low-voltage indication or the alternator warning light (and/or ammeter showing a discharge). First try the standard recovery: check/reset the alternator field circuit breaker, since a tripped breaker is a common, resettable cause. If that fails, reduce electrical load and plan to land before the battery depletes. Since this aircraft's fuel system and ignition (magnetos) are entirely independent of the electrical system, an electrical failure alone does not threaten the engine. *Ref: AFH Ch. 16; Owner's Manual §2-3*

169. How does an in-flight engine fire response differ from an electrical fire? **A:** Engine fire (N8821U checklist): declare an emergency, mixture IDLE CUT OFF, fuel selector OFF, master OFF, airspeed 120 MPH glide, cabin heat/air OFF (to avoid drawing smoke into the cabin through the heater muff); if it persists, increase airspeed to try to blow it out. Don't attempt a restart once secured. Electrical fire: master switch off first (killing the source), then address individual equipment — you don't want to shut off fuel to an engine fire that doesn't exist. *Ref: N8821U checklist; POH Section 3; AFH Ch. 16*

170. What would you do if a door popped open in flight? **A:** Recognize it as a distraction, not an emergency — the airplane flies essentially normally with a door open, just louder and draggier. Maintain aircraft control first, then attempt to close it only if it can be done safely (slow down, open a window/vent to equalize pressure); if it won't close, land as a precaution rather than fixating on it. *Ref: AFH Ch. 16; POH Section 3*

171. What are your notification and reporting responsibilities after an accident or serious incident? **A:** Notify the NTSB "immediately" after an accident or specific serious incidents (flight control malfunction, in-flight fire), and file a written report within 10 days of an accident (7 days for an overdue-aircraft report). Preserve the wreckage and all records until the NTSB releases them, except as needed to protect life/property or in cooperation with law enforcement. *Ref: 49 CFR 830.5, 830.6, 830.10*

D. Emergency Equipment and Survival Gear

172. What emergency equipment is required, and what should you carry beyond the minimum? **A:** Required: an ELT (limited exceptions). A fire extinguisher and first aid kit are commonly required or strongly recommended depending on the operation. Beyond the legal minimum, consider a handheld radio, portable GPS, a red-lens flashlight for night, water, basic first aid, and for remote/mountainous/wilderness terrain, survival gear suited to the environment and season. *Ref: 14 CFR 91.207; AIM 6-2-5*

173. How does an ELT work, and how do you check that it's functioning? **A:** Automatically activates on sudden deceleration and transmits on 121.5/406 MHz to aid search and rescue; a 406 MHz ELT also sends a digital, aircraft-identifying signal to COSPAS-SARSAT for a faster fix. Operational check: within the first 5 minutes after the hour, monitor for no more than 3 audible sweeps — any longer check should be coordinated with ATC/Flight Service to avoid a false alert. *Ref: 14 CFR 91.207; AIM 6-2-5*

X. Night Operations

174. How does human vision change at night, and what techniques compensate? **A:** The eye shifts from cone-based (color, sharp central) vision to rod-based (monochrome, peripheral) vision, taking about 30 minutes to fully adapt and easily disrupted by bright light. Since rods are concentrated away from the retina's center, off-center viewing is more effective at night. Use dim red or white cockpit lighting to preserve dark adaptation and avoid bright lights before/during flight. *Ref: PHAK Ch. 17*

175. What are the most significant night-specific illusions and hazards? **A:** Black hole approach — dark terrain/water with few lights makes the runway appear farther/higher than it is, especially dangerous with no visible horizon. Autokinesis — a stationary light stared at in the dark appears to move. Daytime illusions (featureless terrain, runway width/slope) are also worse at night from fewer references. Mitigated by a stabilized, instrument-cross-checked approach and trusting a VASI/PAPI over visual impression. *Ref: PHAK Ch. 17*

176. What are the night currency and equipment requirements? **A:** Currency: 3 takeoffs and landings to a full stop within the preceding 90 days, between one hour after sunset and one hour before sunrise, to carry passengers at night. Equipment: position/anti-collision lights sunset to sunrise, plus the day-VFR list, a

landing light if operated for hire, and an adequate electrical source. *Ref: 14 CFR 91.209, 61.57(b), 91.205(c)*

177. What special preflight planning considerations apply to a night cross-country? **A:** Verify all aircraft lights work, carry a flashlight with spare batteries, plan the route over lit/populated areas near airports rather than remote/unlit terrain, review terrain/obstruction elevations more conservatively, identify airports with pilot-controlled lighting along the route, and check NOTAMs for lighting outages. *Ref: PHAK Ch. 17; AIM 2-1-8*

178. What additional risk does an engine failure present at night, and how do you mitigate it? **A:** Choosing and judging a landing site is far harder with limited visual cues. Fly toward lit areas/known airports when possible, use the landing light only once committed to the approach (preserves night vision and battery), maintain best-glide speed, and be prepared to land primarily by pitch/airspeed control rather than trying to visually pick a perfect spot. *Ref: AFH Ch. 16; PHAK Ch. 17*

XI. Postflight Procedures

A. After Landing, Parking, and Securing

179. What do you do after landing and clearing the runway? **A:** Complete the after-landing checklist only once clear of the runway, behind the hold-short line — never while still on the runway. Taxi to parking following ground instructions/markings, then complete the engine shutdown and securing checklist (mixture idle cutoff, ignition off, master off, control lock installed, chocks/tie-downs as needed). *Ref: AFH Ch. 2; POH Section 4*

180. What must be logged after a flight, and what's the pilot's postflight responsibility? **A:** Log flight time, landings (including night landings for currency), and any discrepancies noted in flight for maintenance follow-up. The PIC remains responsible for leaving the aircraft in a safe, secure condition appropriate to the parking location and expected weather. *Ref: 14 CFR 43.11, 91.7; AFH Ch. 2*

181. What should you do if you notice new damage or a discrepancy while securing the aircraft? **A:** Note it in the aircraft's discrepancy log or notify the owner/flight school/maintenance directly, and if it affects airworthiness, ensure the aircraft isn't flown again until a mechanic evaluates it — the same PIC responsibility for airworthiness applies at the end of a flight as at the beginning. *Ref: 14 CFR 91.7; AFH Ch. 2*